Career Launcher MBA

QA Algebra-04

Sequences, Series

Question 1

The number of common terms in the two sequences 17, 21, 25,…, 417 and 16, 21, 26,…, 466 is

(1) 78

(2) 19

(3) 20

(4) 77

Question 2

For any natural number n, suppose the sum of the first n terms of an arithmetic progression is $(n + 2n^2)$. If the nth term of the progression is divisible by 9, then the smallest possible value of n is

(1) 8

(2) 7

(3) 9

(4) 4

Question 3

Each of the series S1 = 2 + 4 + 6 +..... and S2 = 3 + 6 + 9 +..... is continued to 100 terms. Find how many terms are identical.

(1) 34

(2) 33

(3) 32

(4) None of these

Question 4

The natural numbers are divided into groups as (1), (2, 3, 4), (5, 6, 7, 8, 9), ….. and so on. Then, the sum of the numbers in the 15th group is equal to

(1) 4941

(2) 6090

(3) 6119

(4) 7471

Question 5

Let a_1, a_2, a_3, a_4, a_5 be a sequence of five consecutive odd numbers. Consider a new sequence of five consecutive even numbers ending with $2a_3$.

If the sum of the numbers in the new sequence is 450, then a_5 is

Question 6

If the square of the 7th term of an arithmetic progression with positive common difference equals the product of the 3rd and 17th terms, then the ratio of the first term to the common difference is

(1) 2 : 3

(2) 3 : 2

(3) 3 : 4

(4) 4 : 3

Question 7

Let a_1, a_2, ... be integers such that $a_1 - a_2 + a_3 - a_4 + ... + (-1)^{n-1} a_n = n$, for all $n \geq 1$.

Then $a_{51} + a_{52} + ... + a_{1023}$ equals

(1) 0

(2) 1

(3) –1

(4) 10

Question 8

If $x_1 = -1$ and $x_m = x_{m+1} + (m + 1)$ for every positive integer m, then x_{100} equals

(1) –5151

(2) –5150

(3) –5051

(4) –5050

Question 9

Consider a sequence of real numbers $x_1, x_2, x_3, \ldots$ such that $x_{n+1} = x_n + n - 1$ for all $n \geq 1$. If $x_1 = -1$ then x_{100} is equal to

(1) 4949

(2) 4850

(3) 4849

(4) 4950

Question 10

If the population of a town is p in the beginning of any year then it becomes 3 + 2p in the beginning of the next year. If the population in the beginning of 2019 is 1000, then the population in the beginning of 2034 will be

(1) $(1003)2^{15} – 3$

(2) $(997)2^{14} + 3$

(3) $(997)^{15} – 3$

(4) $(1003)^{15} + 6$

Question 11

Let the m-th and n-th terms of a geometric progression be 3/4 and 12, respectively, where $m < n$. If the common ratio of the progression is an integer r, then the smallest possible value of $r + n - m$ is

(1) 6

(2) –4

(3) –2

(4) 2

Question 12

Consider the arithmetic progression 3, 7, 11, … and let A_n denote the sum of the first n terms of this progression. Then the value of $\frac{1}{25}\sum_{n=1}^{25} A_n$ is

(1) 442

(2) 415

(3) 455

(4) 404

Question 13

The sum of $1-\frac{1}{6}+\frac{1}{6}\times\frac{1}{4}-\frac{1}{6}\times\frac{1}{4}\times\frac{5}{18}+\ldots$ is :

(1) $\frac{2}{3}$

(2) $\frac{2}{\sqrt{3}}$

(3) $\sqrt{\frac{2}{3}}$

(4) $\frac{\sqrt{3}}{2}$

Question 14

In an infinite geometric progression, each term is equal to twice the sum of all the terms that follow it. If the sum of first two terms is 12, the sum of the entire progression is

(1) 9/2

(2) 27/2

(3) 88/7

(4) 15

Question 15

The sum of 4 + 44 + 444 + ... upto n terms is

(1) $\frac{40}{81}\left(8^n - 1\right) - \frac{5n}{9}$

(2) $\frac{40}{81}\left(8^n - 1\right) - \frac{4n}{9}$

(3) $\frac{40}{81}\left(10^n - 1\right) - \frac{4n}{9}$

(4) $\frac{40}{81}\left(10^n - 1\right) - \frac{5n}{9}$

Question 16

If a, b, c are in G.P. and $a^x = b^y = c^z$, then:

(1) $\frac{1}{x}+\frac{1}{z}=\frac{2}{y}$

(2) $\frac{1}{x}+\frac{1}{z}=-\frac{2}{y}$

(3) $\frac{1}{x}+\frac{1}{y}=\frac{2}{z}$

(4) $\frac{1}{x}+\frac{1}{y}=-\frac{2}{z}$

Question 17

If p, q and r are three unequal numbers such that p, q and r are in A.P., and p, r-q and q-p are in G.P., then p : q : r is equal to:

(1) 1 : 2 : 3

(2) 2 : 3 : 4

(3) 3 : 2 : 1

(4) 1 : 3 : 4

Question 18

Find the sum of following series:

$$\frac{3}{7}+\frac{4}{7^2}+\frac{5}{7^3}+\frac{3}{7^4}+\frac{4}{7^5}+\frac{5}{7^6}+\ldots$$

(1) 180/342

(2) 182/342

(3) 180/343

(4) 182/343

Question 19

Let S denotes the infinite sum $2+5x+9x^2+14x^3+20x^4+\ldots,$

where $|x| < 1$ and the coefficient of x^{n-1} is $\frac{1}{2}n(n+3),(n=1,2,\ldots).$ Then S equals:

(1) $\frac{2-x}{(1-x)^3}$

(2) $\frac{2-x}{(1+x)^3}$

(3) $\frac{2+x}{(1-x)^3}$

(4) $\frac{2+x}{(1+x)^3}$

Question 20

The infinite sum $1+\frac{4}{7}+\frac{9}{7^2}+\frac{16}{7^3}+\frac{25}{7^4}+\cdots$ equals

(1) 27/14

(2) 21/13

(3) 49/27

(4) 256/147

Question 21

Find the sum of the following series:

$$S = \frac{7}{6} + \frac{13}{12} + \frac{21}{20}, \frac{31}{30} + ... \frac{9901}{9900}$$

(1) $\frac{9749}{100}$

(2) $\frac{9849}{100}$

(3) $\frac{9949}{100}$

(4) $\frac{9649}{100}$

Question 22

If the sum of the first ten terms of the series

$\left(1\frac{3}{5}\right)^2+\left(2\frac{2}{5}\right)^2+\left(3\frac{1}{5}\right)^2+4^2+\left(4\frac{4}{5}\right)^2+\cdots$, is $\frac{16}{5}m$, then m is equal to:

(1) 101

(2) 100

(3) 99

(4) 102

Question 23

If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is

(1) 4/3

(2) 1

(3) 7/4

(4) 8/5

Question 24

For each $p > 1$, a sequence $\{A_n\}$ is defined by $A_0 = 1$ and $A_n = p_n + (-1)^n A_{n-1}$ for each $n \geq 1$. For how many integer values of p, 1000 is a term of the sequence?

(1) 8

(2) 7

(3) 5

(4) 4

Question 25

In a certain examination paper, there are n questions. For j = 1,2 …n, there are 2^{n-j} students who answered j or more questions wrongly. If the total number of wrong answers is 4095, then the value of n is

(1) 12

(2) 11

(3) 10

(4) 9

Question 26

If both the sequences x, a1, a2, y and x, b1, b2, z are in A.P. and it is given that y > x and z < x, then which of the following values can {(a1 – a2)/(b1 – b2)} possibly take?

(1) 0

(2) 2

(3) 1

(4) –3

Question 27

From a book, a number of consecutive pages are missing. The sum of the page numbers of these pages is 9808. Which pages are missing?

(1) The page 9808 is missing.

(2) The pages 291 up to 322 are missing.

(3) The page 291 up to and including 322 are missing.

(4) Either (1) or (3)

Question 28

A function $f(x)$ satisfies $f(1) = 3600$ and $f(1) + f(2) + ... + f(n) = n^2 f(n)$, for all positive integers $n > 1$. What is the value of $f(9)$?

(1) 80

(2) 240

(3) 200

(4) 100

Question 29

Consider the sequence of numbers a_1, a_2, a_3, ... to infinity where $a_1 = 81.33$ and $a_2 = -19$ and $a_j = a_{j-1} - a_{j-2}$ for $j \geq 3$. What is the sum of the first 6002 terms of this sequence?

(1) –100.33

(2) –30.00

(3) 62.33

(4) 119.33

Question 30

A marble is dropped from a height of 3 metres onto the ground. After the hitting the ground, it bounces and reaches 80% of the height from which it was dropped. This repeats multiple times. Each time it bounces, the marble reaches 80% of the height previously reached. Eventually, the marble comes to rest on the ground.

What is the maximum distance that the marble travels from the time it was dropped until it comes to rest?

(1) 27 m

(2) 30 m

(3) 15 m

(4) 24 m

Thanks!